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GIS Product Example: West Jordan City Zoning & Emergency Call Analysis

Project Purpose

The planning team wanted to understand which zoning types and areas were generating the highest service costs for the city. To investigate this, they identified a set of specific areas of interest and requested an analysis of emergency call volume. Broken out by year and call type within each area, along with a per-capita rate to allow fair comparison between zones of different populations.

Deliverable

An interactive ArcGIS Dashboard presenting call volume, call type, and per-capita rates by zone and year, with a splash-page explainer so non-GIS staff can navigate the data independently.

https://west-jordan-city.maps.arcgis.com/apps/dashboards/5d0755e8b1f54883b22e8482d01b4936#

Workflow

1. **Define study areas.** Received the planning team's selected areas of interest via PowerPoint boundary sketches.
2. **Build and reconcile polygon features.** Created a polygon feature class matching the requested boundaries, retaining original zoning geometry where it lined up with the planning team's intent, and digitizing new custom polygons from scratch where their PowerPoint boundaries diverged from existing zoning lines. Assign each polygon a Sheet_Nmbr identifier tying it back to the source spreadsheet and slide deck for traceability.
3. **Map and clip call data.** Geocoded/mapped raw emergency call coordinate data and clipped call points to each study area boundary.
4. **Estimate population per zone.** Since the study areas didn't align with existing census geography, apportioned census block population data by land area overlap to produce a population estimate for each custom zone.
5. **Aggregate and normalize.** Counted calls by year and call type within each zone, then converted counts to a per-capita rate (calls per 100 residents) so zones of different sizes and populations could be compared on equal footing.

6. **Build the dashboard.** Assembled the results into an interactive ArcGIS Dashboard, including a splash page with a plain-language explanation of the data and navigation instructions for planning staff.

Data Constraints

Medical call points for 2022 were missing from the source dataset provided. As a result, 2022 medical call counts display as zero in the dashboard, which does not reflect an actual absence of medical calls that year. This was flagged to the Planning team as a known limitation rather than a finding.

Potential Next Step

The dashboard currently shows call volume and rates, not cost. With an average cost-per-call figure by call type, the same zone-level structure could be extended to estimate service cost per area and compare it against revenue generated by each zone, moving the analysis from "which areas generate the most calls" to "which areas are net costs vs. net contributors" to the city.